SIMATS ENGINEERING
ASSESSMENT TOOL 2

ING

COURSE CODE: CSA4307

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.

The invalid field is:

Content-Length: -150

Content-Length must be a non-negative integer. It represents the size of the response body in bytes. Therefore, -150 is invalid.

The other fields are valid:

- HTTP/1.1 200 OK – Valid HTTP status line
- Date – Valid date field
- Content-Type: text/html – Valid content type
- Connection: keep-alive – Valid connection directive

2. Analyze the issue with Content-Length and explain its impact.

The value -150 is invalid because a response body cannot have a negative size.

The Content-Length header informs the browser about the exact number of bytes in the response body. An invalid negative value can cause:

- HTTP protocol parsing errors
- Incorrect interpretation of the response body
- Failure to load the webpage properly
- Connection termination by the browser or server
- Problems with persistent keep-alive connections

Thus, the response header is malformed even though other fields are correct.

3. Determine whether the status code matches the response correctness.

The status code is:

200 OK

This means that the server successfully processed the request. However, the invalid Content-Length makes the response technically incorrect or malformed.

Therefore:

The 200 OK status code indicates successful request processing, but it does not make the invalid header field correct.

4. Suggest corrections for the identified errors.

5. The Content-Length should contain the actual non-negative size of the response body.
6. For example:
7. HTTP/1.1 200 OK

8. Date: Mon, 01 Apr 2026 10:00:00 GMT
9. Content-Type: text/html
10. Content-Length: 150
11. Connection: keep-alive
12. If the actual response body contains 500 bytes, then:
13. Content-Length: 500

14. Conclusion

15. The main error is the negative `Content-Length` value. It must be replaced with the actual non-negative size of the response body to ensure correct HTTP communication.

2: **DOM Structure Analysis**

Consider the following DOM tree representation:

```
<html>
<head>
<title>TestPage</title>
</head>
<body>
<div>
<p>HelloWorld
</div>
</body>
</html>
```

Questions:

1. Missing or improperly closed elements

The element is not explicitly closed.

Incorrect:

HelloWorld

Correct:

HelloWorld

2. How the browser interprets the structure

Browsers use an error-recovery mechanism called HTML parsing error recovery.

When the browser reaches:

while the element is still open, the browser generally automatically closes the paragraph element.

The browser may internally interpret the structure as:

HelloWorld

3. Impact on rendering and layout

In this simple example, the browser will probably display the content correctly. However, invalid or improperly structured HTML can cause:

- Unexpected spacing
- Incorrect nesting
- CSS styling problems
- JavaScript DOM manipulation issues
- Different rendering behavior between browsers
- Accessibility problems in complex pages

4. Corrected valid DOM

```
<html>
<head>
  <title>TestPage</title>
</head>
<body>
  <div>
    <p>HelloWorld</p>
  </div>
</body>
</html>
```

Conclusion

The missing </p> tag should be added explicitly. Although browsers can automatically recover from this error, properly closed elements ensure predictable rendering and a valid DOM structure.

3: GET vs POST in Login System

A login system uses two different methods:

Method URL Example Data Visibility Security Level GET /login?user=admin&pass=1234 Visible in URL Low POST /login Hidden in body Moderate

Questions:

1. Differences in data transmission

GET Method

In GET, data is transmitted as part of the URL.

POST Method

In POST, the data is transmitted inside the HTTP request body:

POST /login HTTP/1.1

The username and password are not displayed in the URL.

Characteristics:

- Data is not visible in the URL.

2. Suitable method for login

POST is more suitable for login authentication.

The login credentials should be sent using POST because the username and password are not exposed in the URL.

For secure transmission, POST must be used together with:

HTTPS

3. Security risks of using GET

Using GET for authentication can expose passwords through:

- Browser history
- Server logs
- Proxy logs
- Browser cache
- Bookmarks
- Screenshots
- Referrer information

For example:

/login?user=admin&pass=1234

Anyone who can access the URL may see the password.

4. Best practices for secure login

The following practices should be followed:

1. Use POST for submitting login credentials.
2. Always use HTTPS/TLS.
3. Never store passwords in plain text.
4. Use secure password hashing algorithms such as bcrypt or Argon2.
5. Use secure session management.
6. Do not store passwords in URLs, logs, or browser history.

Conclusion

POST is more appropriate than GET for login systems, but secure authentication requires both POST and HTTPS, along with secure password storage and session management**.**

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph
<img src="image.jpg">
</body>
```

```
</html>
```

Questions:

1. Identify missing closing tags

The following elements are not explicitly closed:

<h1>Welcome

This is a paragraph

The missing closing tags are:

</h1>

The <img> element does not require a closing tag because it is a void element.

2. Issues with the <img> tag

The given tag is:

<img src="image.jpg">

The main issue is the missing alt attribute.

A better version is:

<img src="image.jpg" alt="Sample image">

The alt attribute is important because:

- It improves accessibility.
- Screen readers can describe the image.
- Alternative text is displayed if the image cannot load.

3. How browsers handle these errors

Modern browsers use error recovery and may automatically infer missing closing tags.

Therefore, the webpage may still render. However, relying on browser error recovery is not recommended because:

- The DOM may not be exactly as intended.

- CSS layout may be affected.
- JavaScript operations may behave unexpectedly.
- Accessibility may be reduced.

4. Corrected HTML Code

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Sample</title>

</head>

<body>

    <h1>Welcome</h1>

    <p>This is a paragraph.</p>

    <img src="image.jpg" alt="Sample image">

</body>

</html>
```

Conclusion

The missing closing tags should be added, and the image should include meaningful alternative text using the alt attribute.

5: Browser-Server Communication Flow

The following sequence describes a web request:

1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request

5. Response is returned
6. Browser renders page

Questions:

1. Interpretation of each step

Step 1: User enters a URL

The browser:

1. Parses the HTML.
2. Creates the DOM.
3. Processes CSS.
4. Executes JavaScript.
5. Calculates the layout.
6. Displays the webpage to the user.

2. Where delays can occur

Delays may occur during:

- DNS lookup
- TCP connection establishment
- TLS/HTTPS handshake
- Network transmission
- Server processing
- Database queries
- Downloading images and other resources
- JavaScript execution

3. Role of DNS

DNS works like an address book for the Internet.

It converts:

Domain Name → IP Address

For example:

www.example.com → Server IP Address

DNS allows users to use easy-to-remember domain names instead of numerical IP addresses.

DNS caching can reduce repeated lookup time.

4. Performance optimizations

DNS Optimization

- Use DNS caching.
- Use reliable DNS servers.
- Reduce unnecessary DNS lookups.

Network Optimization

- Use HTTP/2 or HTTP/3.
- Use persistent connections.
- Use HTTPS efficiently.

Server Optimization

- Use server-side caching.
- Optimize database queries.
- Use load balancing.
- Use a Content Delivery Network (CDN).

File Optimization

- Compress HTML, CSS, and JavaScript.
- Minify CSS and JavaScript files.
- Optimize image sizes.
- Use lazy loading for images.

Browser Optimization

- Reduce unnecessary JavaScript.

- Defer non-critical scripts.
- Cache static resources.

Code of Conduct:

I PINDIBOINA SAI DEEPAK (192411047) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sai Deepak

RUBRICS FOR CALCULATION:

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understandi ng of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand